

Name: BRH-2 **Date:** 04/21/2026 **Class:** WTMA Engineering

MyDesign Portfolio

Element D Initial Template

Element D: Design Concept Generation, Analysis, and Selection

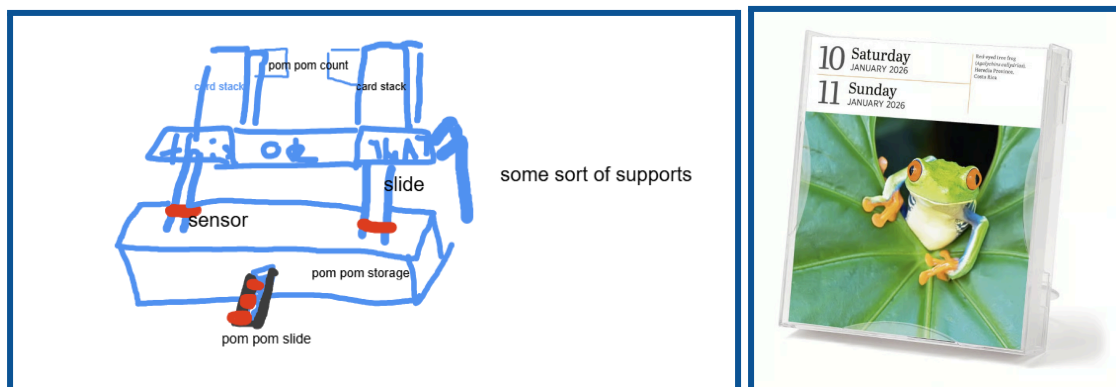
Design Idea Generation (D1)

Design Idea #1 / A

Date: 03/03/2026

Description: This option is a simple design, where we have the card stacks and this or that sign. The pom pom slides has borders so that there were always pom poms ready to be taken, and the slides have sensors on them to count the number of pom poms which were dropped. The pom pom count is somewhere at the top. When the pom poms are dropped and censored, they go into one huge storage bin. Thus there is an infinite supply of pom poms. For the card storage the boxes labeled “this” and “That” would act as those calendars with many pictures and one box underneath

Drawing:

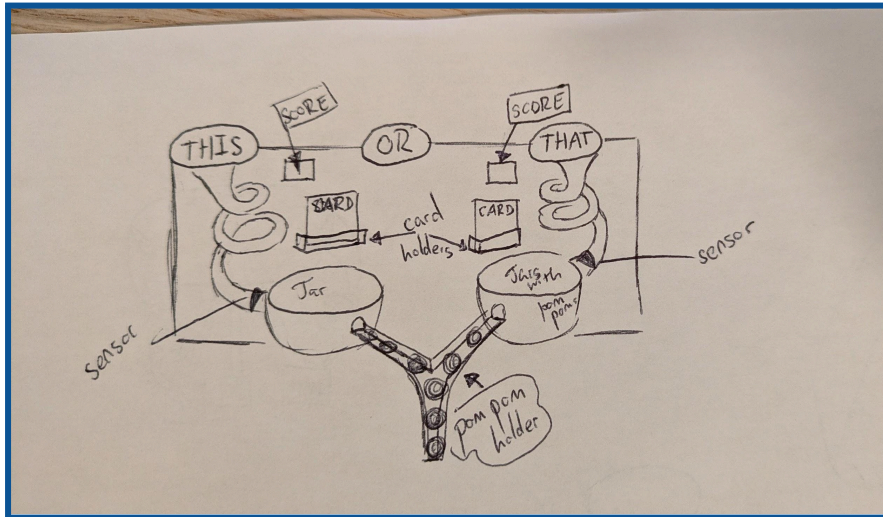


Design Idea #2 / B

Date: 03/03/2026

Description: Here the jars are separate, but the pom poms roll into one holder. The cards are also held in boxes as described above, but this time in the middle of the design.

Drawing:



Design Idea #3 / C

Date: 03/03/2026

Description: When the pom-pom lands in the bin, it presses down a small lever micro switch, which turns on the LED.

What You Need (per side)

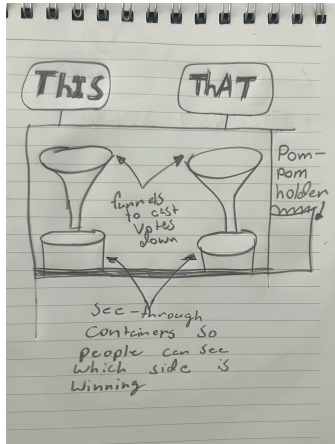
- 1 micro lever switch
- 1 LED strip or 1 bright LED
- 1 resistor (if using single LED)
- 3-4 AA battery pack
- Wires

Setup

1. Mount the micro switch inside the bin at the bottom.
2. Place a thin cardboard flap on top of the switch.
3. When the pom-pom lands, it pushes the flap down.
4. That closes the circuit ... LED lights up.

- ✓ Very simple
- ✓ No coding
- ✓ Reliable

Drawing:

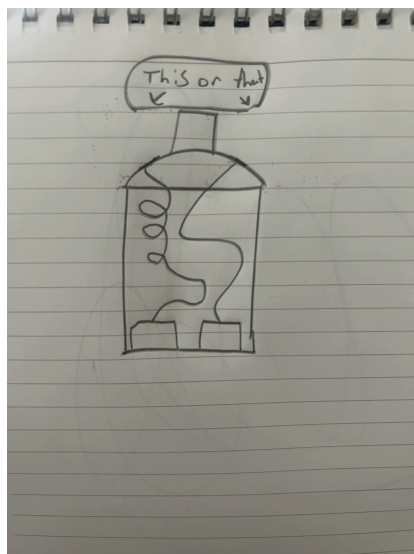


Design Idea #4 / D

Date: 03/03/2026

Description: Kids will decide if they want to cast their vote for "This" or "That," and then they will drop their pom-pom down the corresponding chute. The chute will lead to the clear cup, officially casting their vote in a fun and interactive way.

Drawing:



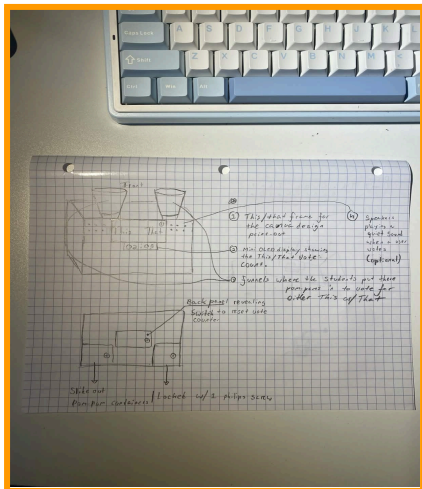
Design Idea #5 / E

Date: 03/03/2026

Description: **User-flow description:**

1. The user places a pom pom in either the "this" or the "that" funnel
2. Two small motion sensors representing the "this" or "that" will detect the movement of a falling pom-pom and add +1 to the "this" or "that" counter displayed on a tiny screen at the front of the device
3. A small randomized sound will play when the vote registered (optional)
4. At the back of the box, the librarian has access to three back panels. One of them allows the librarian to reset the display counter via a switch, while the other two gives them access to the pom-pom storage containers. All of these will be guarded by 1 philips screw.
5. The librarian can swap out the "this or that" prompt by removing and sliding in a new graphic inside a frame located in the front of the contraption.

Drawing:



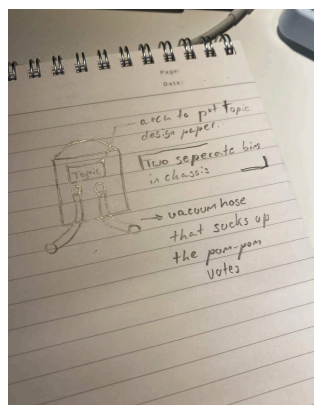
Design Idea #6 / F

Date: 03/03/2026

Description: User-flow description:

The user places the pom-pom in front of the "this" or "that" hose to get sucked up and placed in its respective bin.

Drawing:

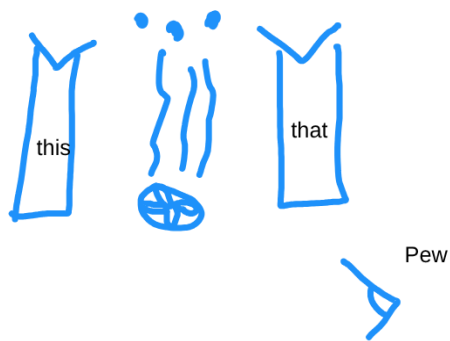


Design Idea #7 / G

Date: 03/03/2026

Description: **Pom-poms float up with the fan and kids shoot them with a nerf dart to whichever tower they want to vote for.**

Drawing:

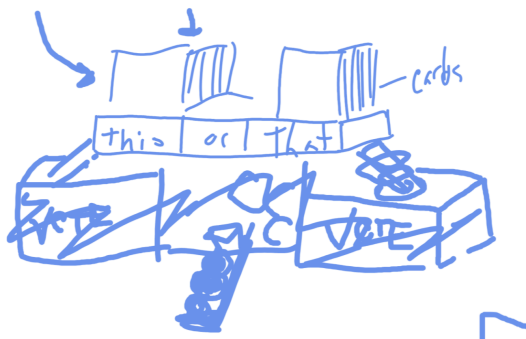


Design Idea #8 / H

Date: 03/03/2026

Description: Pom-poms are dropped through the top towers then going through a slide that leads to a connecting slide then exiting the machine entirely.

Drawing:



Design Idea Comparison and Selection (D1, D2)

Smaller than 11x18 (0.10) Low

Size was important, but less critical than functionality. The machine needed to meet space constraints and be manageable, but being slightly larger wouldn't completely prevent it from working. That's why it was given a lower weight compared to performance-based criteria.

Ability to change the prompt (0.10) Low

This was given a lower weight because, while flexibility is useful, it is not essential to basic operation. The machine can still function with a fixed prompt, so this feature enhances usability but is not critical.

Easy to operate (0.15) Moderate

This was given a moderately high weight because the machine needs to be simple and intuitive for users. If it's confusing or difficult to use, people won't interact with it correctly, which defeats its purpose. However, it's not the most critical factor since the machine can still function even if it takes a little time to learn.

Getting out pom-poms with ease (0.15) Moderate

This was moderately important because smooth operation depends on reliably dispensing the pom-poms. If they get stuck or are difficult to remove, it disrupts the user experience and slows down the process, but it's still secondary to the main purpose of the machine.

Durability (0.20) High

Durability received a higher weight because the machine must withstand repeated use and potential impacts (like drops). If the structure fails, the entire system becomes unreliable. A durable design ensures longevity and consistent performance, making this a key requirement.

Disruption/chaos (0.30) High

This was given the highest weight because the machine is designed to maintain order and function smoothly without causing unnecessary mess or confusion. Instead of encouraging chaos, the goal is to ensure predictable, controlled behavior so users can interact with it easily and consistently. If the machine creates too much disruption (like spilling, jamming, or randomness that interferes with results), it reduces effectiveness and usability. Because this directly impacts the overall success and reliability of the design, it was assigned the greatest importance.

Name: BRH-2 Luca D, Bailey W, Elana B, Jacob L **Date:** 04/21/2026 **Class:** WTMA Engineering

Step 1: choose criteria and weights									
Choose appropriate weights for each criterion, listing each as a decimal									
easy to operate	0.15								
smaller 11x18	0.1								
durability	0.2								

disruption/chaos	0.3								
ability to change the prompt	0.1								
getting out pom-poms w/ ease	0.15								
sum (should = 1)	1								
Step 2: use criteria and weights to compare and select a design (use a weighted decision matrix)									
Luca Drondoe	Weight as decimal	Design Option A	Design Option B	Design Option C	Design Option D	Design Option E	Design Option F	Design Option G	Design Option H
easy to operate	0.15	1	0	1	1	1	1	1	1
smaller 11x18	0.1	1	0	1	0	0	-1	1	0
durability	0.2	0	0	0	0	1	-1	-1	-1
disruption/chaos	0.3	0	1	1	0	0	-1	-1	1
ability to change the prompt	0.1	-1	-1	-1	-1	1	-1	0	0
getting out pom-poms w/ ease	0.15	1	1	0	-1	0	-1	0	1
	total score	0.3	0.35	0.45	-0.1	0.45	-0.7	-0.25	0.4
Elena Bogdanova	Weight as decimal	Design Option A	Design Option B	Design Option C	Design Option D	Design Option E	Design Option F	Design Option G	Design Option H
easy to operate	0.15	1	0	0	1	1	-1	-1	0

smaller 11x18	0.1	0	0	1	1	0	-1	0	0
durability	0.2	-1	0	1	1	1	0	-1	0
disruption/chaos	0.3	0	0	0	0	0	-1	-1	0
ability to change the prompt	0.1	1	1	0	-1	1	0	0	1
getting out pom-poms w/ ease	0.15	1	0	-1	-1	0	-1	0	1
	total score	0.2	0.1	0.15	0.2	0.45	-0.7	-0.65	0.25
Jacob Lindsey	Weight as decimal	Design Option A	Design Option B	Design Option C	Design Option D	Design Option E	Design Option F	Design Option G	Design Option H
easy to operate	0.15	1	0	-1	1	0	0	1	
smaller 11x18	0.1	0	-1	1	0	0	-1	0	
durability	0.2	-1	0	0	0	1	0	-1	
disruption/chaos	0.3	1	0	-1	0	0	-1	-1	
ability to change the prompt	0.1	0	1	0	-1	0	0	0	
getting out pom-poms w/ ease	0.15	1	1	-1	-1	-1	-1	-1	
	total score	0.4	0.15	-0.5	-0.1	0.05	-0.55		
Bailey Walker	Weight as decimal	Design Option A	Design Option B	Design Option C	Design Option D	Design Option E	Design Option F	Design Option G	Design Option H
easy to operate	0.15	1	0	1	1	1	0	0	1
smaller 11x18	0.1	0	0	1	0	0	1	-1	0

durability	0.2	-1	0	0	0	1	-1	0	-1
disruption/chaos	0.3	1	1	1	1	1	-1	-1	1
ability to change the prompt	0.1	0	1	0	0	0	-1	0	0
getting out pom-poms w/ ease	0.15	0	0	0	-1	0	-1	-1	1
	total score	0.25	0.4	0.55	0.3	0.65	-0.65		

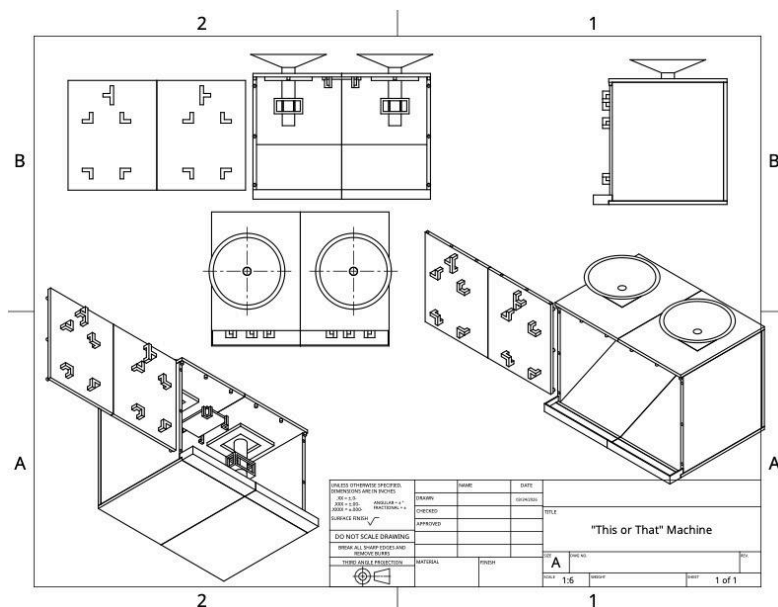
Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1 | Prototype Iteration 1

Date: 4/8/26

Sketch:



Manufacturing Methods:

Step 1: Wiring the arduino, reset button, and sensors

Step 2: programming the functionality of the reset button and sensors

Step 3: 3D printing and assembling chassis

Supply List and Costs:

Item	Cost each	# needed	Total
Half-sized bread board	\$4.95	1	\$4.95
16mm Panel Mount Momentary Pushbutton	\$0.95	1	\$0.95
Adafruit METRO 328 w/ headers	\$17.50	1	\$17.50
Premium Male/Male Jumper Wires	\$1.95	1	\$1.95
Assembled Standard LCD 16x2 w/ headers	\$10.95	1	\$10.95
IR break beam sensors	\$2.95	2	\$2.95
3D filament	15	3	45

MyDesign Portfolio - Element D Initial Template

Name: _____		Date: _____	Class: _____
Total Cost	84.25		

Plan of Action (D3)

Gantt Chart 03/23 - 03/ 28

